Traditional Chinese Medicine Toxicity Prediction by Heterogeneous Network

Supplementary File

1. Dataset

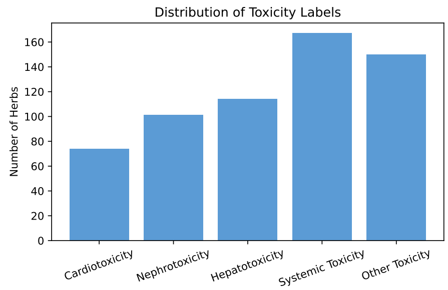
1.1. Dataset construct

To construct the herb toxicity dataset, we compiled a comprehensive list of herbs from authoritative sources such as Chp (2020) [1], HERB [2], SymMap v2 [3], TCM-Bank [4], and HIT 2.0 [5]. For each herb, toxicity-related information was manually retrieved and curated from major scientific databases and literature repositories, including Google Scholar (`https://scholar.google.com`), CNKI (China National Knowledge Infrastructure, `https://www.cnki.net/`), VIP (China Science and Technology Journal Database, `https://www.cqvip.com/`), WanFang (Wanfang Data, `https://www.wanfangdata.com.cn/index.html`), PubMed (`https://pubmed.ncbi.nlm.nih.gov/`), and WOS (Web of Science, `https://www.webofscience.com/wos/`). All collected data were manually screened, annotated, and standardized, ensuring the consistency and reliability of herb names and toxicity information across sources.

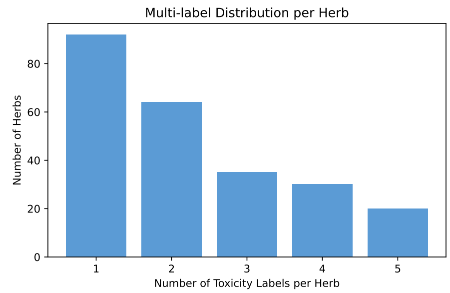
We defined five toxicity labels: cardiotoxicity, nephrotoxicity, hepatotoxicity, systemic toxicity, and other toxicity, based on clinical relevance and literature prevalence. For each herb, the presence or absence of a toxicity type was determined by manually reviewing the collected literature and database entries. Our final dataset consists of 252 herbs, each potentially associated with more than one toxicity labels. In total, 74 herbs are labeled as cardiotoxic, 101 as nephrotoxic, 114 as hepatotoxic, 167 as systemically toxic, and 150 as having other toxicities.

1.2. Dataset analysis

Figure S1 presents a statistical overview of the constructed multi-label toxicity dataset. As depicted in Figure S1(a), the distribution of toxic labels among 252 collected herbs is notably imbalanced. Systemic toxicity and other toxicity are observed



(a) Distribution of toxicity labels among the herb dataset.



(b) Distribution of the number of toxicity labels assigned to each herb.

Figure S1: Statistical overview of the multi-label toxicity dataset. (a) The frequency of each toxicity label among the collected 252 herbs, highlighting the imbalanced distribution across different toxicity types. (b) The distribution of toxic labels per herb, demonstrating the prevalence of multi-label toxicity within the dataset.

to be the most frequently assigned labels, occurring in 167 and 150 herbs, respectively, whereas cardiotoxicity is comparatively rare, with only 74 herbs. Nephrotoxicity and hepatotoxicity are present in 101 and 114 herbs, respectively. This uneven distribution suggests that certain toxicity types are more commonly reported or studied, which may introduce challenges for model training and evaluation.

Figure S1(b) further highlights the pronounced multi-label characteristics of the dataset. While a portion of herbs are annotated with a single toxic label, the majority are associated with two or more toxic types, with some herbs assigned up to five distinct labels. This prevalence of multi-label annotations reflects the inherent complexity of herb toxicity and underscores the necessity for predictive models to effectively capture the interactions and dependencies among multiple toxicity outcomes. The dominance of multi-label instances further increases the challenge of the prediction task, demanding approaches that are capable of addressing such complexity.

Figure S2 displays the co-occurrence heatmap of toxic labels within our dataset. The results demonstrate that certain toxicity types tend to co-occur at relatively high frequencies. Notably, hepatotoxicity and nephrotoxicity are jointly annotated in 76 herbs, indicating a strong association between these two toxicity types. Additionally,

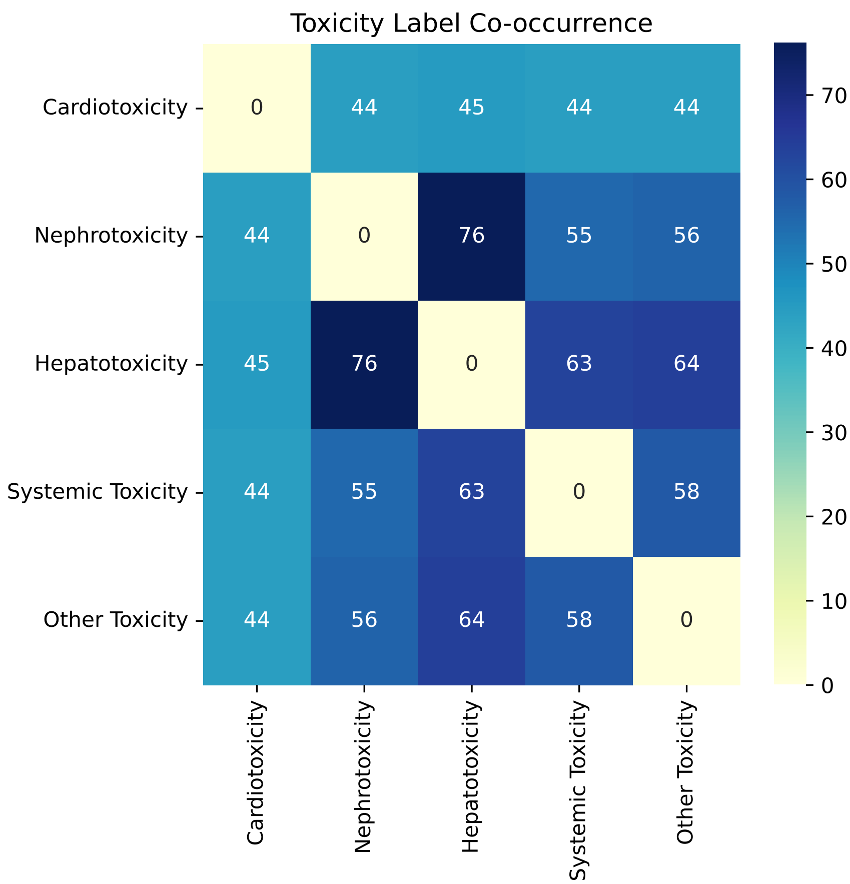


Figure S2: Co-occurrence heatmap of toxic labels of herbs.

systemic toxicity frequently co-exists with nephrotoxicity (55 herbs) and other toxicity (58 herbs), reflecting potential interdependencies or shared risk factors among these categories. Such co-occurrence patterns may reflect potential biological correlations or shared mechanisms underlying multiple toxic responses. The presence of frequent co-occurrences, together with label imbalance and prevalent multi-label associations, highlights the inherent complexity of the dataset and may increase the challenges associated with developing accurate and generalizable toxicity prediction models of herbs.

2. Ablation study

To further study the contribution factors of `HerbToxNet`, we introduce five variants: (i) **w/o HAN** directly uses the properties and efficacy data of herb as features without constructing heterogeneous graph and the meta-path, but combined with contrastive learning and weighted label fusion strategies for prediction. (ii) **w/o PE** randomly initializes the herb node features in the heterogeneous graph instead of using pharmacological property and efficacy data as initial features. (iii) **w/o CL** removes the contrastive learning module with dynamic coefficients in the training phase. (iv) **w/o δ** replaces the dynamic coefficient δ_{ij} with the toxicity label similarity β_{ij}. (v) **w/o WLF** only uses the MLP for prediction, without incorporating the weighted label fusion strategy.

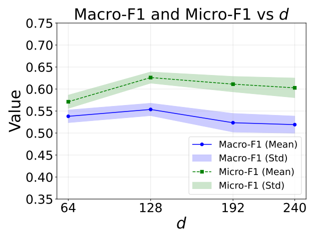
Table S1 shows the results of `HerbToxNet` and its degenerated variants. Overall, `HerbToxNet` significantly outperforms its variants across all evaluation metrics, highlighting the importance of each designed module in the full model. Specifically: (i) `HerbToxNet` clearly outperforms w/o HAN, this indicates HAN effectively models the complex relationships between herbs, ingredients, and targets, learns more discriminative herb representations by meta-path, thus providing a more reliable foundation for toxicity prediction. (ii) `HerbToxNet` gains better results than w/o PE, suggesting that pharmacological property and efficacy data provide valuable information for herb toxicity prediction, despite their inherent incomplete and insufficiency. These data supplement the model with domain knowledge, helping the model to better understand the toxic characteristics of herbs. (iii) `HerbToxNet` is superior to w/o CL, owing to the contrastive learning which pulls together herbs with similar toxic labels and pushes apart dissimilar ones, this significantly enhances the robustness and accuracy of the model. (iv) `HerbToxNet` outperforms w/o δ, which validates that the hyperparameter t can adjust β and allow for more flexible quantification of semantic similarity between herbs, thereby rewarding contrastive learning. (v) `HerbToxNet` is better than w/o WLF, suggesting the weighted label fusion strategy effectively leverages knowledge from other similar herbs, further improving prediction accuracy by fusing the toxic labels from similar herbs and by alleviating the scarce data issue.

We observe that the performance of w/o CL is the poorest across all metrics, underscoring the critical role of the contrastive learning module in enhancing discriminative representation of herbs. The variant w/o HAN performs only slightly better, highlighting the importance of the meta-path mechanism in capturing the complex semantic associations among herbs, ingredients, and targets. In contrast, although w/o PE and w/o WLF achieve relatively higher scores, their performance remains significantly inferior to `HerbToxNet`, suggesting that the integration of pharmacological property and efficacy information, along with the weighted label fusion strategy, substantially contributes to model prediction. The performance of w/o δ falls between that of w/o HAN and w/o PE, indicating that dynamic coefficient adjustment has a non-negligible impact on modulating the contrastive learning process, albeit to a lesser extent. Overall, these results demonstrate that each component in `HerbToxNet` is important, and their synergy is crucial to achieve the state-of-the-art performance.

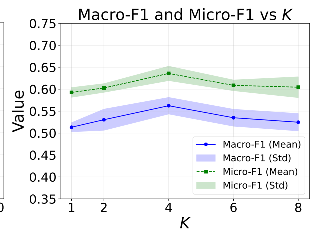
Table S1: Prediction results of `HerbToxNet` and its variants.

	Macro-F1	Micro-F1	AvgAUC	1−OneError
w/o HAN	0.4934±0.0357	0.5352±0.0218	0.5003±0.0245	0.5335±0.0378
w/o PE	0.4766±0.0312	0.5435±0.0304	0.4854±0.0153	0.5475±0.0370
w/o CL	0.4366±0.0292	0.5305±0.0262	0.4851±0.0168	0.5500±0.0177
w/o δ	0.5087±0.0342	0.5411±0.0216	0.5252±0.0206	0.5486±0.0152
w/o WLF	0.5269±0.0317	0.6131±0.0233	0.5363±0.0313	0.5536±0.0511
`HerbToxNet`	**0.5652± 0.0161**	**0.6247±0.0172**	**0.6003± 0.0141**	**0.5587±0.0485**

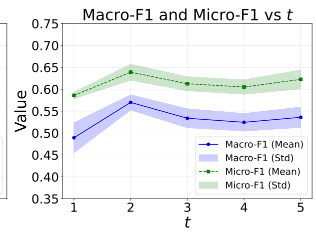
Note: The best results are highlighted in **bold font**, with statistical significance checked by student t-test at 95% level.



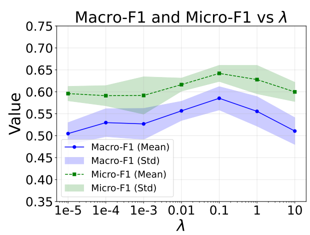
(a) node embedding dimension d



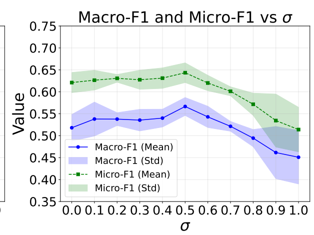
(b) attention head number K



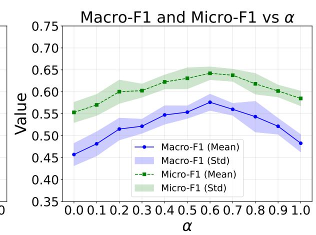
(c) adjustment parameter t



(d) loss function balancing parameter λ



(e) weighted label fusion threshold σ



(f) fusion proportion parameter α

Figure S3: Results of parameter sensitivity analysis. (a), (b), (c) and (d) show the Macro-F1 and Micro-F1 of `HerbToxNet` under different input values of node embedding dimension d, head number K, adjustment parameter t, and loss function balancing parameter λ, respectively; (e) and (f) show the Macro-F1 and Micro-F1 of `HerbToxNet` under different input values of the weighted label fusion threshold σ and fusion proportion parameter α, respectively.

3. Parameter sensitivity analysis

`HerbToxNet` is implemented by Python program language. For the model training, We used an Nvidia GeForce RTX 3090 (24G) GPU that hosted on a Ubuntu 22.04.1 LTS machine with Intel Xeon Gold 6248R CPUs and 512GB memory. The version of CUDA is 11.7 and that of PyTorch is 1.13.1.

In this experimental environment, we systematically investigated the impact of several important parameters: (i) d determines the representation dimensions of nodes such as herbs, ingredients, and targets in the heterogeneous graph; (ii) K denotes the number of attention heads in the HAN, which allows the model to jointly attend to information from different representation subspaces at different positions. (iii) Contrastive learning dynamic coefficient adjustment parameter t is used to adjust the toxic-

ity label similarity β_{ij}, which affects the calculation of the dynamic coefficient δ_{ij}. (iv) Loss function balancing parameter λ balances the contrastive learning loss $\mathcal{L}^{con}$ and the asymmetric classification loss $\mathcal{L}^{asl}$. (v) Weighted label fusion threshold σ selects similar herbs with higher similarity to supplement the labels. (vi) Fusion proportion parameter α controls the weights of predictions from MLP and weighted label fusion. We test d within $\{64, 128, 192, 240\}$, K within $\{1, 2, 4, 6, 8\}$, t within $\{1, 2, 3, 4, 5\}$, λ within $\{10^{-5}, 10^{-4}, 10^{-3}, 10^{-2}, 10^{-1}, 10^{0},$
$10\}$, both σ and α within $\{0, 0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9, 1.0\}$ on our dataset. When testing each parameter, the other parameters are fixed at their optimal values and the results are shown in Figure S3.

The node embedding dimension d plays a pivotal role, `HerbToxNet` achieves the optimal performance when $d = 128$, suggesting that this embedding dimension effectively captures the intricate associations among herbs, ingredients, and targets within the heterogeneous network. In contrast, when $d = 64$, the model exhibits a significantly lower Macro-F1 score, indicating that a lower embedding dimension limits the model's ability to capture the complex relationships among nodes. However, when $d \geq 128$, its performance even falls below that observed at $d = 64$. This suggests that a excessively large embedding dimension may introduce noise, while also incurring substantially higher computational costs. Consequently, we select $d = 128$ as the optimal embedding dimension to ensure effective representation of nodes.

The number of attention heads K has a significant impact on the performance of `HerbToxNet`. Experimental results demonstrate that the model achieves its best performance when $K = 4$, emphasizing the effectiveness of employing multi-head attention to capture diverse semantic patterns from heterogeneous node interactions. When $K = 1$, the single-head attention mechanism lacks sufficient capacity to distinguish diverse semantic patterns, resulting in a noticeable drop in prediction performance. As $K > 4$, the performance starts to decline, likely due to excessive attention heads dispersing informative signals, thereby weakening the overall feature representation. Therefore, we choose $K = 4$ as the optimal configuration to ensure sufficient expressiveness without introducing noise or instability.

The dynamic coefficient adjustment parameter t plays a critical role in the optimization process of contrastive learning. Experimental results indicate that the model achieves optimal performance when $t = 2$. At this value, the model effectively balances the amplification of high-similar samples and the penalization of low-similar ones. When $t = 1$, the dynamic coefficient reduces to δ_{ij}, which insufficiently amplifies the contributions of high-similar samples. This limitation hinders the model's ability to clearly distinguish between the similarities of different samples. Conversely, $t > 4$ excessively penalizes low-similar samples. While this adjustment further emphasizes high-similar samples, it causes the model to disproportionately focus on these samples, potentially neglecting other valuable information. This overemphasis can lead to a decline in overall model performance. As such, we set $t = 2$ to enhance the contributions of high-similar samples.

The balance parameter λ plays a pivotal role in the model training process. Experimental results indicate that the model achieves optimal performance when $\lambda = 0.1$, as this setting establishes a reasonable balance between the contrastive learning loss and the multi-label classification loss. As λ increases, the model's performance initially improves but subsequently declines, suggesting that an excessive emphasis on the contrastive learning loss diminishes the optimization effect of the multi-label classification loss on the final classification task. Conversely, when the contribution of the contrastive learning loss is insufficient, the model fails to fully leverage the similarity information among samples. Such, we set $\lambda = 0.1$ to effectively utilize the contrastive learning strategy while simultaneously addressing the final classification objective.

The weighted label fusion threshold σ plays a critical role in filtering similar neighbor samples during the retrieval process. Experimental results show that the model achieves optimal performance when $\sigma = 0.5$. Specifically, when σ is too small, the model tends to include an excessive number of noisy samples. This results in the inclusion of many toxic labels with weak associations to the target herb, thereby disrupting the model's learning process. Conversely, when σ is too large, the number of similar neighbor samples is significantly reduced, limiting the model's ability to effectively leverage information from these neighbors during training. Therefore, we set $\sigma = 0.5$ as the similarity threshold to select relevant similar samples and filter out noise.

The fusion proportion parameter α determines the relative contribution of the target sample's labels and those of its similar samples in the weighted label fusion strategy. Experimental results indicate that the model achieves optimal performance when $\alpha = 0.6$. This suggests that moderately increasing the weight for toxic labels of neighbor samples effectively enhance the model's predictive capability. When $\alpha < 0.6$, the limited influence of neighbor herbs hinders effective label fusion. Meanwhile, $\alpha > 0.6$ causes the model to overly depend on neighbors, weakening the target sample's own representation and reducing the performance. Thus, we set $\alpha = 0.6$ to balance the toxic labels induced from the target herb and its neighbor herbs.

Based on the parameter sensitivity analysis, we set the parameter values as follows: $d = 128$, $K = 4$, $t = 2$, $\lambda = 0.1$, $\sigma = 0.5$, and $\alpha = 0.6$. The general training-related parameter settings of our method, as well as those of all baseline methods, are summarized in Table S2.

Table S2: Parameter settings of `HerbToxNet` and all baseline methods.

	Learning rate	Training epoch	Input size	Dropout rate	Optiminzer
ZF-LightGBM	0.01	100	252×364	–	–
QSAR-ANN	0.001	200	252×364	–	Adam
QSAR-SVM	–	Until convergence	252×364	–	Internal
EI-MS-RF	–	300	252×364	–	–
DeepDILI	0.0001	100	252×364	–	Adam
R-E-GA	–	150	252×364	–	–
TIRPnet	0.001	200	252×364	–	Adam
ML-PRDF	–	–	252×364	–	–
MulGRaL	0.001	200	252×364	0.3	Adam
MLC-CNN	0.001	100	252×364	0.5	Adam
`HerbToxNet`	0.001	200	252×364	0.2	Adam

Note: "–" indicates that the parameter is not applicable to the corresponding method.

4. Toxicity prediction of herbs

In this case study, we use `HerbToxNet` to predict and analyze the toxicity of canonical herbs: *Chai Hu*, *He Shou Wu*, *Ba Dou*, *Fu Zi*, and *Bi Ma Zi* in real-world scenarios. *Chai Hu* and *He Shou Wu* are widely recognized for their potential hepatotoxicity and included to assess the model's performance in predicting liver-related adverse effects. *Ba Dou* and *Fu Zi*, classified as extremely "cold" and "hot" herbs respectively, are examined to explore their toxicity associations with liver, kidney, and cardiac systems. Additionally, we incorporated the herb *Bi Ma Zi*, which was classified as "ping" (neutral in nature and generally considered non-toxic) in traditional theories, to study whether the model predictions are consistent with the traditional view that neutral herbs are essentially safe. In the experimental setup, besides these five herbs, all other herbs in the original dataset are used as the training set, while the five selected herbs are treated as a separate test set for final prediction. This setup effectively evaluates the generalization capability of `HerbToxNet` on completely unknown samples. The predicted toxicities are then compared with known toxic labels, and relevant literature is retrieved from PubMed to verify whether the predictions align with current research findings. This experiment further assesses the scientific validity and authenticity of `HerbToxNet`.

4.1. `HerbToxNet` *can identify toxicity of canonical hepatotoxic herbs*

As shown in Table S3, `HerbToxNet` accurately predicts three major toxicities of *Chai Hu*: hepatotoxicity, nephrotoxicity, and systemic toxicity. The primary categories of active ingredients in *Chai Hu* include saikosaponins, volatile oils, flavonoids, and polysaccharides, which are responsible for both its pharmacological and toxicological properties. Saikosaponins have been shown to induce hepatotoxicity, including elevated liver enzymes and hepatocyte necrosis, likely due to inhibition of hepatic drug-metabolizing enzymes, especially in individuals with impaired liver function [6]. The predicted systemic toxicity can be attributed to mucosal irritation caused by volatile oils and saponins, leading to gastrointestinal symptoms such as nausea, vomiting, or diarrhea [6]. For nephrotoxicity, long-term administration of a certain dose of total saikosaponins purified by ethanol elution from *Chai Hu* has been shown to cause significant cumulative toxicity in rats, with the primary target organs of toxic damage being the liver and kidneys [7]. For *He Shou Wu*, `HerbToxNet` also produces accurate toxicity predictions, including hepatotoxicity and nephrotoxicity, along with a possible systemic toxicity signal. *He Shou Wu* contains ingredient categories such as anthraquinones, stilbenes, polysaccharides, and phospholipids. Although it is widely used for its antioxidant and hepatoprotective effects, excessive or prolonged use are associated with severe liver injury, including hepatitis, liver failure, and even liver transplantation [8]. The predicted nephrotoxicity aligns with evidence that anthraquinones can cause tubular damage and interstitial fibrosis in animal models [9]. Although direct evidence for systemic toxicity is limited, reports of allergic reactions, including immune-mediated symptoms such as rash and pruritus, suggest the model can capture low-frequency or context-dependent systemic toxic effects [10].

4.2. `HerbToxNet` *can reveal toxicity risks in extremely cold or hot herbs*

To further investigate the relationship between extreme herb properties and toxicity, we analyzed representative extremely "cold" and "hot" herbs *Ba Dou* and *Fu Zi*, respectively. According to traditional theory, such herbs exhibit potent bioactivity and may cause overstimulation or disruption of organ function, contributing to their toxic potential. According to the results shown in Table S3, `HerbToxNet` successfully identified multiple toxicities of *Ba Dou*, including nephrotoxicity, hepatotoxicity, systemic toxicity (digestive system toxicity) and other toxicity (genetic toxicity). The active ingredients *croton oil* and *phorbol esters* of *Ba Dou* possess significant toxic effects. The hepatotoxicity of *Ba Dou* is primarily attributed to the direct damage of its ingredients to hepatocyte membranes and mitochondria, potentially resulting in cell necrosis and elevated liver enzymes [11]. The digestive system toxicity stems from the strong irritant effects of *croton oil*, which can compromise intestinal barrier integrity and trigger mucosal inflammation [12]. Its genetic toxicity is attributed to *phorbol esters*, which promote DNA damage and chromosomal aberrations via protein kinase C activation, contributing to carcinogenic processes [11]. Although literature on its nephrotoxicity is relatively limited, animal studies have shown that high doses of *Ba Dou* cause renal congestion and tubular degeneration, indicating kidney damage [13].

Similarly, `HerbToxNet` accurately predicts the cardiotoxicity of *Fu Zi*. The primary toxic ingredient of *Fu Zi* is *aconitine alkaloids*, which can rapidly act on cardiac sodium channels when overdosed or improperly processed, leading to arrhythmia, bradycardia, or even sudden cardiac death [14]. Overall, these results indicate that `HerbToxNet` can make accurate and reliable toxicity prediction of extremely "cold" or "hot" herbs.

Table S3: Results of toxicity prediction for *Chai Hu*, *He Shou Wu*, *Ba Dou*, *Fu Zi*, *Bi Ma Zi*

Herb Name	Rank	Toxicity	Prediction	Known	PubMed ID
Chai Hu	1	Hepatotoxicity	√	√	21749378
	2	Nephrotoxicity	√	√	34309413
	3	Systemic Toxicity	√	√	21749378
	4	Other Toxicity	×	×	-
	5	Cardiotoxicity	×	×	-
He Shou Wu	1	Hepatotoxicity	√	√	25648693
	2	**Systemic Toxicity**	√	×	**19873791**
	3	Nephrotoxicity	√	√	32426606
	4	Other Toxicity	×	×	-
	5	Cardiotoxicity	×	×	-
Ba Dou	1	Other Toxicity	√	√	35619875
	2	Nephrotoxicity	√	√	32943947
	3	Hepatotoxicity	√	√	35619875
	4	Systemic Toxicity	√	√	30717554
	5	Cardiotoxicity	×	×	-
Fu Zi	1	Cardiotoxicity	√	√	30261585
	2	Nephrotoxicity	×	×	-
	3	Hepatotoxicity	×	×	-
	4	Systemic Toxicity	×	×	-
	5	Other Toxicity	×	×	-
Bi Ma Zi	1	Hepatotoxicity	√	√	12565756
	2	Nephrotoxicity	√	√	12565756
	3	Other Toxicity	√	√	16278363
	4	Systemic Toxicity	√	√	16278363
	5	Cardiotoxicity	√	√	16278363

4.3. `HerbToxNet` *can uncover toxicity risks in neutral-property herbs*

To examine the traditional assumption that "ping" (neutral-property) herbs are inherently non-toxic, we selected *Bi Ma Zi*, a representative ping herb, for in-depth case analysis. According to classical TCM theory, herbs with a neutral property are considered balanced and safe. As shown in Table S3, despite being classified as a neutral herb in classical TCM texts such as the "Bencao Gangmu", where neutral-property herbs are traditionally considered to possess balanced qualities and safe with low toxicity, modern toxicological studies reveal a different picture for *Bi Ma Zi*. This herb contains highly toxic ingredients, primarily *ricin* and *ricinine*, which can induce severe systemic toxic effects upon ingestion. Specifically, *ricin* exhibits pronounced hepatotoxicity, leading to hepatocellular necrosis and elevated liver enzymes, and nephrotoxicity, characterized by acute tubular injury and renal failure [15, 16]. Additionally, *ricin* can cause cardiotoxicity, manifesting as hypotension and arrhythmias; neurotoxicity, including headaches and altered consciousness; gastrointestinal toxicity, such as severe vomiting, diarrhea, and hemorrhage; and significant immunoallergic reactions [16]. Based on these findings, *Bi Ma Zi* is selected as a representative neutral herb to critically assess the traditional assumption that neutral-property herbs are inherently non-toxic. The aim is to uncover potential toxic risks and to provide scientific evidences that may refine and complement the classical TCM classification system.

4.4. `HerbToxNet` *can uncover the relationship between herb properties and toxicity*

To examine the correlation between predictions of `HerbToxNet` and traditional TCM theory, we also investigate the relationship between TCM properties and toxicity. The toxicity of several herbs, including *Fu Zi*, *He Shou Wu*, *Ba Dou*, *Bi Ma Zi*, and *Chai Hu*, are closely associated with their pharmacological properties. *Fu Zi*, with its hot nature, acrid taste, and heart, kidney, and spleen tropism, is known for cardiotoxicity and neurotoxicity due to toxic alkaloids. *He Shou Wu*, characterized by bitter flavor and liver and kidney tropism, is frequently linked to hepatotoxicity. *Ba Dou*, bearing extreme acrid-hot properties and stomach and large intestine tropism, causes strong gastrointestinal and systemic toxicity. Although *Bi Ma Zi* is traditionally considered neutral and sweet, its liver and kidney meridian affinity corresponds with significant hepatotoxic and nephrotoxic effects. *Chai Hu*, a cold and bitter herb targeting the liver meridian, is also associated with liver toxicity, consistent with its traditional classification. These cases demonstrate that the core TCM properties of nature, flavor, and meridian tropism provide valuable indicators for assessing herb toxicity risk within the TCM paradigm. The toxicity predictions generated by `HerbToxNet` for these canonical herbs are consistent with the toxicity profiles. This suggests that `HerbToxNet` can effectively leverage the information encoded in traditional herb properties (nature, flavor, meridian tropism) to predict herb toxicity.

5. Biointerpretability analysis

In the task of predicting herb toxicity, the interactions between herb ingredients and their targets play a pivotal role in mediating pharmacological effects and toxic responses. A single herb typically contains multiple bioactive ingredients that can act on diverse targets, triggering cascades of signaling events that influence cellular functions and organ systems, ultimately manifesting as complex toxicological phenotypes. To enhance the biointerpretability of `HerbToxNet`, we constructed a multi-level explanatory framework encompassing the "herb–ingredient–target–biological pathway–toxicity" axis, using *Chai Hu* as a representative case study for biological validation.

Based on the "Model interpretability analysis" subsection, we first identified the herbs associated with *Chai Hu* through the HIH and HTH meta-paths. We then conducted further analysis on these herbs. In terms of ingredients, we found that these herbs, as confirmed by HIH meta-path, share common classes of secondary metabolites, including triterpenoid saponins, flavonoids, phenolic compounds, volatile oils, and bitter alkaloids. By integrating known toxicity information, we ultimately identified four toxic ingredients, denoted as *Saikosaponin A*, *Saikosaponin D*, *Wogonin*, and *Eugenol*. We extracted 49 distinct targets that are implicated in hepatotoxicity, nephrotoxicity, and systemic toxicity from the herbs confirmed by HTH meta-path. These targets are potentially involved in toxicity mechanisms triggered by interactions with the identified toxic ingredients. Subsequently, to establish the ingredient–target–toxicity associations, we conducted an extensive literature review and cross-referenced the results with existing toxicity databases. As a result, we constructed a tripartite association table centered on the toxic ingredients, linking them to their corresponding targets and toxicity types.

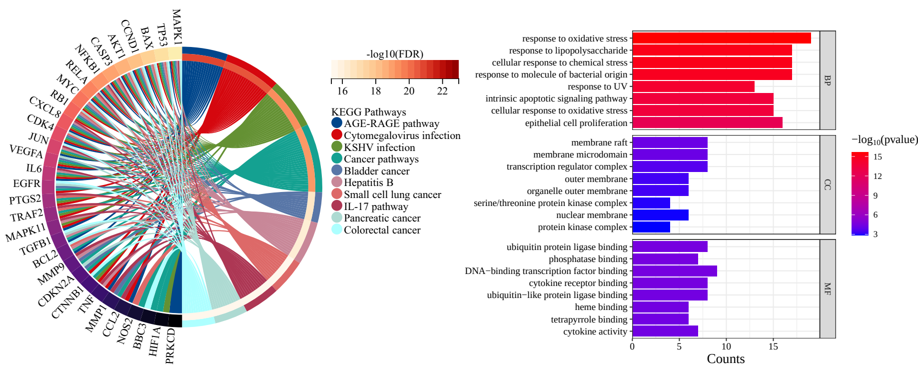
As shown in Table S4, the toxic ingredients identified from *Chai Hu* display heterogeneous toxicological characteristics, as reflected by their diverse target profiles and associations with various toxicity types. *Saikosaponin A* and *Saikosaponin D*, two triterpenoid saponins, are associated with several apoptosis and proliferation related

Table S4: *Chai Hu* ingredients–targets–toxicity relationship

Ingredient Class	Ingredient Name	Target (Gene Symbol)	Toxicity Type
Triterpenoid saponin	Saikosaponin A	CASP3, RB1, BCL2, BAX, MYC, CDKN2A	Hepatotoxicity, Nephrotoxicity
Triterpenoid saponin	Saikosaponin D	JUN, TGFB1, TNF, RELA, IL6, NR3C1, BCL2, NFKB2A, MYC, TP53, CDK4	Hepatotoxicity, Nephrotoxicity
Flavonoid	Wogonin	MAOA, MAOB, NOAS2, PTGS2, ABCC2, CYP1B1	Systemic toxicity
Phenolic compound	Eugenol	NFKB1, BAX, CASP3, TP53, PTGS2, BCL2, NOS2, CSK3B, VEGFA, CYP1A2, HIF1A, CTNNB1, CYP2C19, RELA, PRKCD, MMP1, MCL1, CCL2, CDK9, CCND1, BBC3, CKDN1A, IL6, CXCL8, KDR, AKT1, PTGS1, JUN, MMP9, CDKM1B, MAPK11, TNF, ERN1, EGFR, SPP1, CDK4, TRAF2, MAPK1	Hepatotoxicity

targets, such as *CASP3*, *BCL2*, *MYC*, and *TP53*, suggesting their potential to disrupt cell survival and homeostasis in hepatic and renal tissues, thereby contributing to both hepatotoxicity and nephrotoxicity. *Wogonin*, classified as a flavonoid, interacts with targets including *MAOA*, *PTGS2*, and *CYP1B1*, which are known to participate in inflammatory responses and metabolic activation, indicating a mechanism of systemic toxicity likely mediated by oxidative stress and immune dysregulation. *Eugenol*, a phenolic compound, exhibits the broadest target spectrum, with 38 distinct targets implicated, such as *IL6*, *TNF*, *NFKB1*, *EGFR*, and *MAPK1*, many of which are central to inflammatory signaling, cellular stress responses, and tissue injury. This extensive target involvement underscores *Eugenol*'s strong association with hepatotoxicity and highlights its potential role as a major toxic ingredient in *Chai Hu*.

To further elucidate the biological functions and involved toxicological pathways, we performed KEGG pathway enrichment and Gene Ontology (GO) annotation analyses on these toxicity-associated targets. As shown in Figure S4, KEGG pathway enrichment analysis revealed that the top significantly enriched pathways include the AGE-RAGE pathway, Cytomegalovirus infection, KSHV infection, and a range of cancer-associated pathways such as cancer pathways, pancreatic cancer, and bladder cancer. Many of these pathways are tightly linked to chronic inflammation, oxidative stress, and dysregulated apoptosis processes that are known mediators of tissue injury and organ toxicity. For example, the activation of AGE-RAGE axis is known to induce reactive oxygen species (ROS) production and NFKB activation, leading to sustained inflammatory responses and fibrotic remodeling, which are key pathological features of liver and kidney injury [17]. Similarly, IL-17 signaling and hepatitis B pathway enrichment suggest immune-driven cytotoxicity and pro-inflammatory cytokine release, both of which have been implicated in hepatic inflammation and nephroinflammation [18].



(a) KEGG pathway enrichment

(b) GO annotation

Figure S4: Enrichment analysis results of toxicity-associated targets, (a) KEGG pathway enrichment, (b) Gene Ontology annotation.

GO annotation results also corroborate these findings. In the Biological Process (BP) category, significantly enriched terms such as response to oxidative stress, cellular response to chemical stress, response to lipopolysaccharide, and intrinsic apoptotic signaling pathway, indicate that the identified targets are heavily involved in the cellular response to toxic insults. These biological processes represent core mechanisms of hepatocyte and renal tubular cell injury, where oxidative stress and mitochondrial dysfunction often lead to apoptosis or necrosis. The enriched terms like epithelial cell proliferation may reflect compensatory regeneration following tissue injury. In the Cellular Component (CC) category, the enrichment of membrane raft and organelle outer membrane suggests the localization of toxic target proteins at key subcellular compartments involved in signal transduction and stress sensing. Enriched Molecular Function (MF) terms, such as cytokine receptor binding, ubiquitin protein ligase binding, and heme binding, are functionally related to inflammatory signaling, protein degradation, and metabolic enzyme activity, respectively. Notably, heme binding and tetrapyrrole binding are relevant to cytochrome P450 activity, implicating impaired xenobiotic metabolism as a contributor to drug-induced liver and kidney damage. These results suggest that the toxic targets of *Chai Hu* are embedded in a complex signaling network that governs immune activation, oxidative stress responses, apoptosis, and metabolic detoxification. Toxic ingredients in *Chai Hu* may interfere with these targets, thereby disrupting critical cellular processes and ultimately contributing to hepatotoxicity, nephrotoxicity, and systemic toxicity.

In summary, this biointerpretability analysis highlights the capability of `HerbToxNet` to uncover mechanistically meaningful and biologically plausible pathways underlying herb-induced toxicity. By integrating pharmacological and toxicological evidence along with the "herb–ingredient–target–toxicity" axis, `HerbToxNet` not only enhances the model interpretability but also provides a valuable basis for risk assessment of herbs. This case study of *Chai Hu* exemplifies how multi-level evidences can be systematically linked to elucidate potential toxicological mechanisms, thereby supporting safer and more informed use of herbs.

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